

Element L

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Content of Recommendations:

If our group had another six months to work on this project, we would have done a few things differently. We succeeded this far into the project, and we have realized that we did not achieve all of our goals.

Capture data:

If the project were extended, capturing exact data would be our number one priority. We tried using a syringe to see how high the water would rise if it were pulled by 10cm. Unfortunately, the syringe was not very useful for calculating exactly how high the water would rise.

Adding Electrical Components:

Having electrical components would make calculating and storing data possible. A solution to this would be applying the arduino kit to this device, or totally constructing a different device. The user would still inhale through the device. The sensor would be able to calculate exact data. Unfortunately when the water rises, no data is recorded.

Comparing our project to a real device:

We would have tried to obtain an actual MIP respiratory device to compare how our project differs from an actual device. If we had more time, calculating would be the biggest thing we would have wanted to achieve. Our group has seen many ideas from other devices to another,

and we created something no one has ever seen before. Because of the nature of this product, we could not work off of anything. The idea of using water and negative pressure came from a barometer experiment from a science teacher on youtube. Our group decided to take that route because one of Drexel's requests was to make something that was inexpensive.

L2: If we could continue with the project, the first step would be to replace the PVC we are currently using with a new one to implement the slit design instead of the gradual design. Doing that would allow us more ease in collecting data. We can make the height of each slit coordinate with a certain level of difficulty, allowing people to level up the difficulty as needed with ease. Also, this regulates the levels in a more concrete way than the continuous slit, which can change with any minor adjustment in the PVC pipe itself.